Audio Equipment and Systems

Bachelor's Degree in Audiovisual Systems Engineering

Introduction. Basic concepts

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Department of Information
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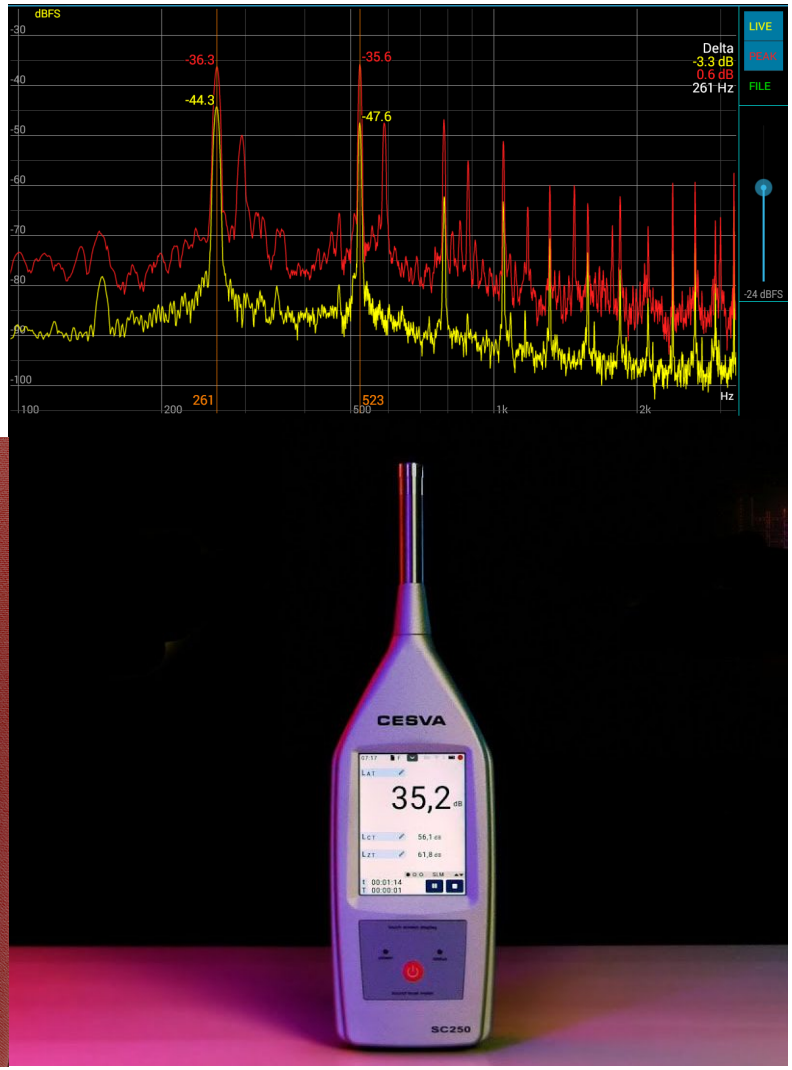
Introduction. Basic concepts

- Sound Energy, Intensity, Power, and Pressure
- SIL, SPL and SWL
- Influence of distance
- Weighted levels
- Signal levels
- Adding levels



Energy, Intensity, Power and Pressure

Sound Energy: The capacity of a sound source to do work. As the source vibrates, it does work in the process of moving the air molecules around itself (the energy is transferred to the air molecules). Energy is usually denoted by a capital "E" and the unit of energy is the joule (J).

Sound Power: How much energy is transferred in every second. In a sound source it describes how much energy is expended per second in moving the surrounding air molecules. We will denote power by a capital "W", the unit of power is the watt (W = J/s).

Sound Intensity: The amount of sound energy that passes through a particular area in one second (Power/area). Sound intensity is usually denoted by a capital "I" and the unit of intensity is W/m^2 or Wm^{-2}.

Sound Pressure: As the sound source vibrates backward and forward, it applies cicles of force to the air molecules, pushing them together (compression) and leaving regions of fewer molecules (rarefaction). The moving molecules then collide with adjacent ones and pass on the energy. The sound pressure is the amount of force that is applied by the moving molecule per unit area. We will use "p" to denote Sound pressure, which unit is the pascal (Pa) with 1Pa = 1 N/m^2.

SPL (Sound Pressure Level): Is the logarithmic representation of Sound Pressure referenced to the threshold of hearing, $20.10^{-6}Pa$

$$SPL = 20log(\frac{p_1}{p_0}) = 20log(\frac{p_1}{20.10^{-6}})$$ (Example: 1Pa =94dB SPL)

Relationship between Sound Pressure and Sound Intensity: At T=20º c=344m/s, the characteristic acoustic impedance is 400 rayls

$$I = \frac{p^2}{Z_a} = \frac{p^2}{\rho c} = \frac{p^2}{400}$$

Intensity threshold $$I_0 = \frac{p_0^2}{\rho c} = \frac{(20.10^{-6})^2}{400} = 10^{-12} W/m^2$$

SIL (Sound Intensity Level): Is the logarithmic representation of Sound Intensity referenced to the intensity threshold, $10^{-12}W/m^2$

$$SIL = 10log\frac{I_1}{I_0} = 10log\frac{I_1}{10^{-12}}$$

Sound Level changes: $\Delta L_p = 20log\frac{p_2}{p_1}$ In free field, spherical radiation +6dB every time we double the Pressure.

$$I \propto p^2 \rightarrow \Delta L_I = 10log\frac{I_2}{I_1} = 10log(\frac{p_2}{p_1})^2 = 20log\frac{p_2}{p_1} = \Delta L_p$$

SIL, SPL and SWL

SWL (Sound Power Level): Is the logarithmic representation of Sound Power referenced to the power reference, $10^{-12}W$ (see *)

$$SWL = 10log\frac{W_1}{W_0} = 10log\frac{W_1}{10^{-12}}$$

Sound Power Level changes: $\Delta L_W = 10log\frac{W_2}{W_1}$ In free field, spherical radiation, +3dB every time we double the Power.

Relationship between SIL, SPL and SWL: $I = \frac{W}{A} = \frac{W}{4\pi r^2}$ in spherical radiation

$$L_p = L_W - |10log\frac{Q}{4\pi r^2}| \quad ; \text{ Q = directivity factor}$$

At 1m, free field, spherical radiation, T=20º, c=344m/s, $Z_0 = 400$ Pa.s/m :

$$L_p = L_W - |10log\frac{1}{4\pi}| = L_w - 11$$

$$SIL = SPL = SWL - 11$$

In free field, spherical radiation, T=20º, c=344m/s, $Z_0 = 400$ Pa.s/m :

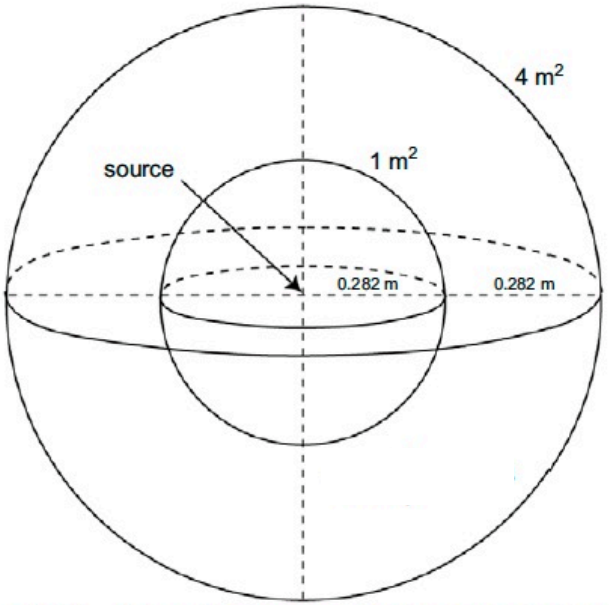
$$L_p = L_W \leftrightarrow \frac{Q}{4\pi r^2} = 1 \rightarrow r = \sqrt{\frac{Q}{\pi r}} = 0.2821m$$

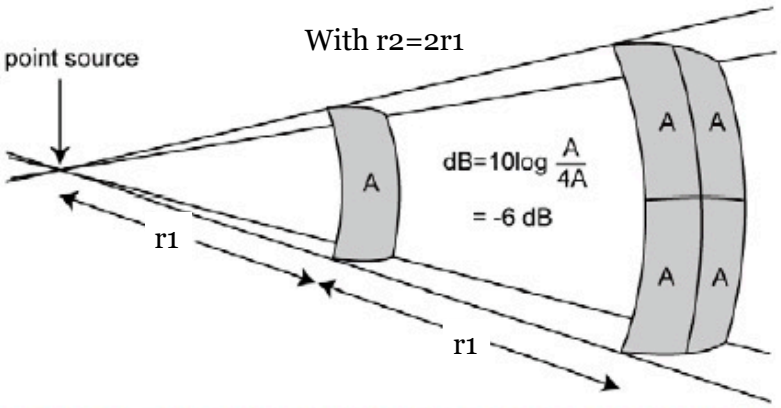
SIL = SPL = SWL at r = 0.2821m (distance where the propagation area is $1m^2$) $I = W \rightarrow I_0 = W_0 = 10^{-12}$ *

Sound pressure (p) at radial distance from a source with known sound power (W): $I = \frac{W}{A} = \frac{QW}{4\pi r^2} = \frac{p^2}{Z_a} = \frac{p^2}{\rho c}$

$$p = \sqrt{\frac{Q\rho cW}{4\pi r^2}}$$

Influence of distance. Spherical radiation



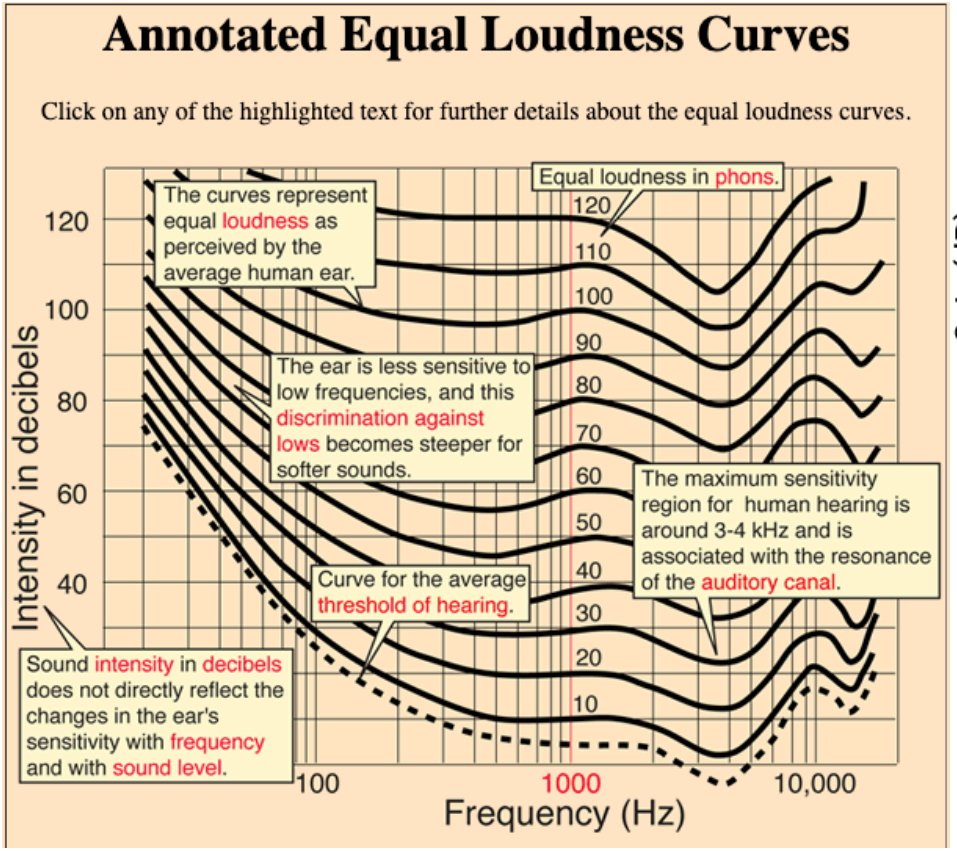


$$\Delta L = L_2 - L_1 = 10log\frac{I_2}{I_0} - 10log\frac{I_1}{I_0} = 10log\frac{\frac{W}{4\pi r_2^2}}{I_0} - 10log\frac{\frac{W}{4\pi r_1^2}}{I_0} = 10log\frac{r_1^2}{r_2^2}$$

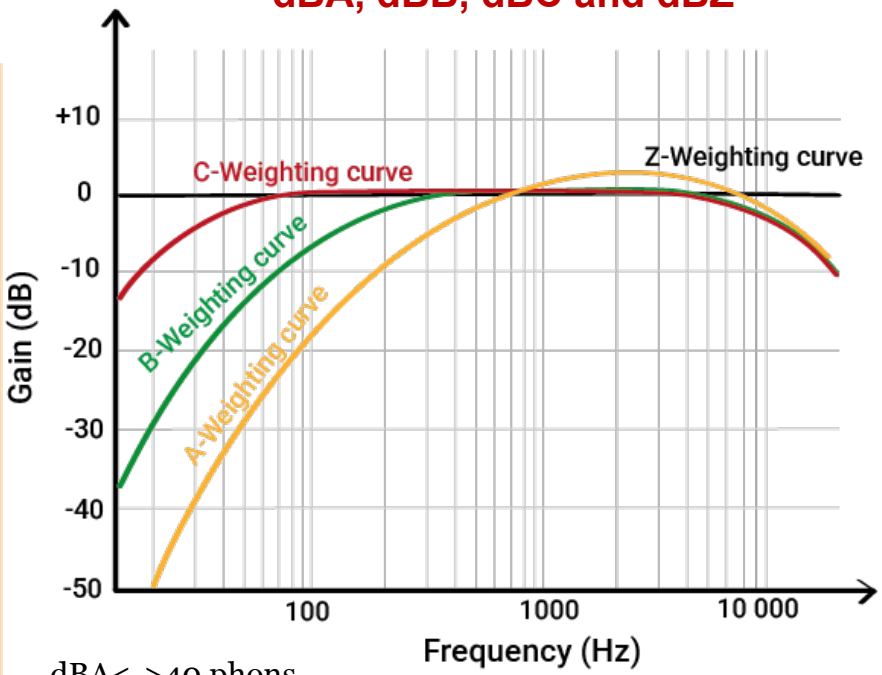
Inverse square law $\Delta L = 20log\frac{r_1}{r_2}$

In free field, spherical radiation, -6dB every time we double the distance

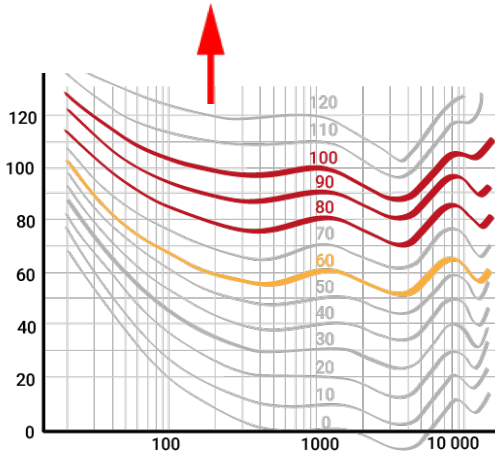
Weighted levels



dBA, dBB, dBC and dBZ



dBA<->40 phons
dBB<->70 phons
dBC<->100 phons



http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/eqloud.html

dBA are typically used to measure low levels ~60 phons (speech, music, etc...)
dBC are typically used to measure high levels ~100 phons (noise, machines, etc...)

https://ateliercrescendo.ac/understanding-acoustic-weighting-curves-where-does-dba-comes-from/

Signal levels

Pressure

$$dBSPL = 20log\frac{P}{P_0} = 20log\frac{P}{20.10^{-6}}$$

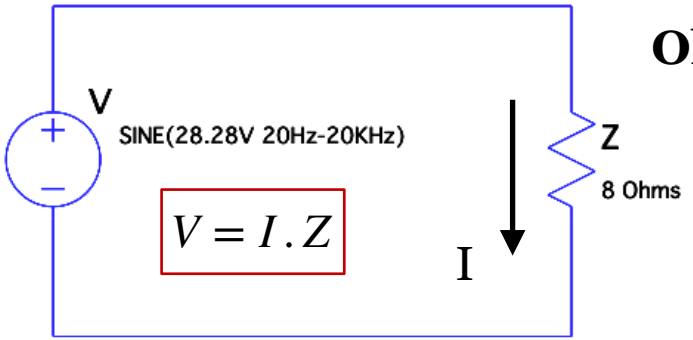
+ 6dB every time we double the Pressure

Voltage

$$dBu = 20log\frac{V}{v_0} = 20log\frac{V}{0.775}$$

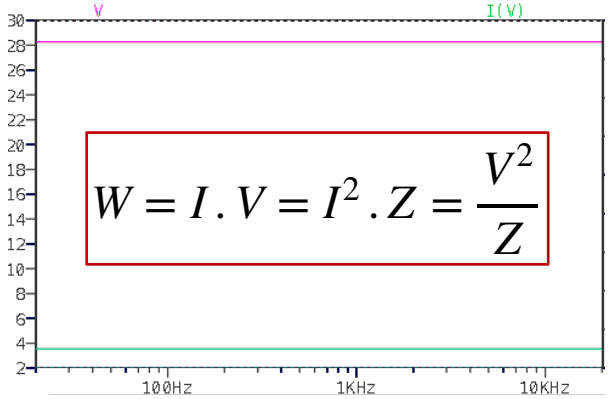
$$dBV = 20log\frac{V}{v_0} = 20logV$$

+ 6dB every time we double the Voltage



Ohm's Law
(Linear)

$$V = I.Z$$



$$W = I.V = I^2.Z = \frac{V^2}{Z}$$

Power

$$dBW = 10log\frac{W}{W_0} = 10logW$$

$$dBm = 10log\frac{W}{W_0} = 10log\frac{W}{10^{-3}}$$

+ 3dB every time we double the Power

Adding levels

Not coherent sources: sum of sound intensities (Ex: reverberation, uncorrelated)

$$L_i = 10log(\frac{I_i}{I_o}) \rightarrow \frac{I_i}{I_0} = 10^{\frac{Li}{10}}$$

$$L_{tot} = 10log(\sum_{i=1}^{N} 10^{\frac{L_i}{10}})$$

+ 3dB every time we double the level

Coherent sources: sum of sound pressures (Ex: clusters, identical signals)

$$I = \frac{p^2}{Z_0} \rightarrow L_i = 10log(\frac{p_i^2 Z_o}{Z_o p_o^2}) = 10log(\frac{p_i^2}{p_o^2}) = 20log(\frac{p_i}{p_o}) \rightarrow \frac{p_i}{p_0} = 10^{\frac{Li}{20}}$$

$$L_{tot} = 20log(\sum_{i=1}^{N} 10^{\frac{L_i}{20}})$$

+ 6dB every time we double the level

Bibliography

BOOKS

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LINKS

- Matlab calculators: *http://www.mcsquared.com/java.htm*
- Calculators: http://www.sengpielaudio.com
- Sound. The wave equation: https://www.feynmanlectures.caltech.edu/I_47.html
- Equal Loudness: *http://hyperphysics.phy-astr.gsu.edu/hbase/Sound/eqloud.html*
- Weighted levels: *https://ateliercrescendo.ac/understanding-acoustic-weighting-curves-where-does-dba-comes-from/*
- Sound Physics: https://www.britannica.com/science/sound-physics

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